

Commutative rings as split Grothendieck rings

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Abstract

Every commutative unital ring is the split Grothendieck ring of a small additive rigid symmetric monoidal category in which every object is self-dual. This answers Question 15 of Levy. We first construct categories of polynomial objects and finite-pattern maps whose coproduct and product relations are prescribed by nonnegative polynomials. A unification argument supplies finite limits, while a development invariant determines the object isomorphisms. Passing to additive spans produces a rigid category; categorical dimension identifies its split Grothendieck ring with the original Burnside ring. An object representing the negative of the unit then allows us to extract any prescribed ring from the resulting polynomial extension. We also describe the construction for $\mathbb{F}_4$, where a square-zero endomorphism has trace one.

1 Introduction

For an additive category $\mathcal{C}$, its split Grothendieck group $K_0^\oplus(\mathcal{C})$ is generated by the isomorphism classes of objects, with relations

$$[A \oplus B] = [A] + [B].$$

If $\mathcal{C}$ is symmetric monoidal and its tensor product is additive in each variable, tensor product makes $K_0^\oplus(\mathcal{C})$ a commutative ring. Levy asks which commutative rings admit such a realization when every object of $\mathcal{C}$ is dualizable [3, Question 15].

Theorem 1.1. *For every commutative unital ring R , there exists a small additive rigid symmetric monoidal category $\mathcal{C}_R$ and an isomorphism of unital rings*

$$K_0^\oplus(\mathcal{C}_R) \cong R.$$

The category can be chosen so that every object is self-dual.

The construction first realizes nonnegative polynomial relations. The extra variables in the following statement record the objects that witness these relations by coproduct decompositions.

Theorem 1.2. *Let I, J be sets and let $P_j \in \mathbb{N}[x_i \mid i \in I]$ be nonzero polynomials, for $j \in J$. There is a small additive rigid symmetric monoidal category $\mathcal{A}_{\mathbf{P}}$, all of whose objects are self-dual, with*

$$K_0^\oplus(\mathcal{A}_{\mathbf{P}}) \cong \mathbb{Z}[x_i, u_j \mid i \in I, j \in J] / (P_j \mid j \in J),$$

where the u_j are additional indeterminates.

To prove Theorem 1.2, we construct a category $\mathcal{E}(\mathbf{P})$ whose objects are polynomial expressions in sets X_i, U_j , with specified decompositions $U_j \cong U_j \amalg P_j(\mathbf{X})$. Its morphisms act by finitely many constructor patterns. Two properties of this category are needed. Finite limits permit composition of spans; they follow by solving the constructor equations defining equalizers in Section 2. The exact presentation of its object semiring requires control of isomorphisms. In Section 3, a finite-pattern inverse forces each branch of an isomorphism to preserve its sorted labels, and development of patterns gives the desired semiring presentation.

The additive span category of $\mathcal{E}(\mathbf{P})$ is rigid. In [Section 4](#), we compute its categorical dimension and obtain an inverse to the map on Grothendieck rings induced by the identity on objects. This proves [Theorem 1.2](#). For a general ring presentation, one additional variable z , subject to $1 + z = 0$, converts every relation to a nonnegative polynomial relation. The resulting category has Grothendieck ring $R[\mathbf{u}]$. In [Section 5](#), the object of class -1 supplies a stabilization inside the full subcategory of objects whose classes lie in R , which proves [Theorem 1.1](#).

The development argument is modeled on Blass's description of explicit bijections between polynomial expressions in finite binary trees [[2](#), Theorem 3]. His relation is $X = 1 + X^2$; here we use the independent relations $u_j = u_j + P_j(\mathbf{x})$ and include maps that repeat or omit labels in order to obtain finite limits. The passage from spans to duality is the usual compact-closure construction [[4](#), Corollary 5.8]; the comparison of Grothendieck rings is proved directly below. Finally, [Section 6](#) describes an endomorphism over $\mathbb{F}_4$ whose square is zero and whose trace is one, relating the construction to the trace-zero condition of [[3](#), Definition 3 and Theorem 4].

Conventions

All rings are commutative and unital, and $\mathbb{N} = \{0, 1, 2, \dots\}$. An additive category is a preadditive category with finite biproducts. For a small category $\mathcal{E}$ with finite coproducts, $K_0^\Pi(\mathcal{E})$ denotes the group completion of its monoid of object isomorphism classes under coproduct. When finite products distribute over finite coproducts, this is a ring, called the Burnside ring; the corresponding semiring before group completion is the Burnside semiring. A category is called *lex* extensive if it has finite limits and finite coproducts that are disjoint and stable under pullback. Products indexed by empty sets are terminal objects, and empty coproducts are initial objects.

2 Finite-pattern maps and finite limits

Fix sets I and J and nonzero polynomials

$$P_j(\mathbf{x}) \in \mathbb{N}[x_i \mid i \in I], \quad j \in J.$$

Every polynomial has finite support. Choose a finite set A_j and a map $\alpha_j : A_j \rightarrow \mathbb{N}^{(I)}$ such that

$$P_j(\mathbf{x}) = \sum_{a \in A_j} \mathbf{x}^{\alpha_j(a)}.$$

Different elements of A_j may have the same image under α_j ; these repetitions account for the coefficients of P_j .

For each $i \in I$, choose a countably infinite set X_i . Regard these sets as distinct sorts, and put

$$D_{j,a} = \prod_{i \in I} X_i^{\alpha_j(a)_i}, \quad D_j = \prod_{a \in A_j} D_{j,a}, \quad U_j = \mathbb{N} \times D_j.$$

Only finitely many factors in the displayed product are nontrivial. The constructors

$$\begin{aligned} s_j : U_j &\longrightarrow U_j, & s_j(n, d) &= (n + 1, d), \\ c_{j,a} : D_{j,a} &\longrightarrow U_j, & c_{j,a}(d) &= (0, (a, d)) \end{aligned}$$

are injective, have pairwise disjoint images, and give a decomposition

$$U_j = s_j(U_j) \amalg \prod_{a \in A_j} c_{j,a}(D_{j,a}). \tag{2.1}$$

Every element of U_j has a unique expression $s_j^n c_{j,a}(d)$. We call n its height. Since $P_j \neq 0$, every U_j is infinite. Write

$$\mathcal{S} = \{X_i \mid i \in I\} \amalg \{U_j \mid j \in J\}$$

for the set of sorts; lower-case x_i and u_j denote the corresponding formal variables.

2.1 Patterns and their parameter objects

A *polynomial object* is an element of

$$\mathbb{P} = \mathbb{N}[x_i, u_j \mid i \in I, j \in J].$$

Its underlying set $|A|$ is obtained by interpreting addition as tagged disjoint union, multiplication as Cartesian product, and the variables as their respective sorts. Fix representations of monomials by finite lists of factors and of coefficients by finite sets of tags. Thus each $|A|$ is a specified set; reordering factors and regrouping tags gives the usual structural bijections.

Definition 2.1 (Linear patterns). A label has a sort in $\mathcal{S}$. An elementary term of sort X_i is a label of that sort. An elementary term of sort U_j has one of the forms

$$s_j^n(\lambda), \quad s_j^n c_{j,a}(\lambda_{i,r}),$$

where $n \geq 0$, the first label has sort U_j , and the labels in the second expression have the sorts of the factors of $D_{j,a}$.

A *linear pattern* in a monomial object is a tuple of elementary terms, one per factor, in which each label occurs exactly once in the whole tuple. A linear pattern in a polynomial object also specifies a tagged summand. Write $L(p)$ for the labels of a pattern p ; these are also called its frontier labels.

The *parameter object* of p is the monomial

$$M_p = \prod_{\lambda \in L(p)} S_\lambda,$$

where S_λ is the sort of λ , interpreted in this expression as its polynomial variable. Evaluation defines an injective function $\iota_p : |M_p| \rightarrow |A|$; injectivity follows by cancelling the constructors. Its image is the *cylinder* of p . A finite family of linear patterns is a *pattern partition* if its cylinders are pairwise disjoint and cover $|A|$. The empty family is a partition of 0.

A cylinder imposes particularly simple conditions on its coordinates. A term $s_j^n(\lambda)$ requires height at least n , whereas a term $s_j^n c_{j,a}(\lambda_{i,r})$ requires height exactly n and seed tag a . All remaining parameters vary independently. In particular, a linear pattern imposes no equality between distinct parameters.

Definition 2.2 (Liberal patterns). For a finite set L of sorted labels, a *liberal pattern over L* is a tagged tuple of elementary terms using labels from L , with each label allowed to occur any number of times, including zero.

Definition 2.3 (Finite-pattern maps). A function $f : |A| \rightarrow |B|$ is a *finite-pattern map* if there are a finite pattern partition $(p_r)_{r \in F}$ of A and liberal patterns q_r in B over $L(p_r)$ such that

$$f(\iota_{p_r}(\rho)) = q_r(\rho)$$

for every assignment ρ of values to these labels. Maps are identified when their underlying functions agree.

The coordinate projections, diagonals, tagged injections, and constructors are finite-pattern maps. So are the structural bijections for products and coproducts and the inverse to the constructor decomposition (2.1). More generally, every pattern partition gives mutually inverse finite-pattern functions

$$\prod_{r \in F} M_{p_r} \longleftrightarrow A. \quad (2.2)$$

The forward function evaluates each pattern. On the cylinder of p_r , the inverse records its tag r and its tuple of labels. This also specifies precisely what is meant by extracting the parameters of a pattern.

2.2 Development and composition

To *develop* a label of sort U_j once, replace it by the successor case and the finitely many seed cases in (2.1), using fresh labels in each case. Developing a label in a linear pattern partitions its cylinder into finitely many cylinders. The standard depth- n partition of U_j consists of

$$s_j^n(\lambda), \quad s_j^r c_{j,a}(\lambda_{i,k}) \quad (0 \leq r < n, a \in A_j).$$

Products and tagged unions give the standard depth- n partition of every polynomial object. Any finite family of pattern partitions has a common such refinement: choose n strictly larger than all depths appearing in the patterns, and develop each tail $s_j^r(\lambda)$ a further $n - r$ times. Seed terms already exposed require no development. Patterns that differ only by a renaming of labels are identified here.

Lemma 2.4 (Preimages of cylinders). *Let $f : |A| \rightarrow |B|$ be a finite-pattern map, and let C be the cylinder of a linear pattern q in B . Then $f^{-1}(C)$ is a finite disjoint union of cylinders. These cylinders can be chosen as part of a pattern partition of A so that, for each corresponding source pattern p ,*

$$\iota_q^{-1} \circ f \circ \iota_p : |M_p| \longrightarrow |M_q|$$

is given by a tuple of elementary terms. In particular, the inclusions of the parameter objects and the displayed extraction maps are finite-pattern maps.

Proof. On a branch in a description of f , membership in C tests the output summand, finitely many heights, and finitely many seed tags. Develop every U -sorted source label to a common depth N larger than the depths occurring in the output terms and in q .

Each test is constant on a resulting branch. Indeed, an output term already ending in a seed has a fixed height and seed tag. For an output term $s_j^a(\lambda)$, a target tail of depth b asks whether the height of λ is at least $\max(b - a, 0)$; a target seed of depth b asks for a specified tag at height $b - a$, or is impossible if $b < a$. The chosen development decides all these conditions. Repeated labels in the output cause no additional tests, since the target pattern is linear.

Retain the branches on which all tests succeed. On each such branch, cancel the fixed successors and seed constructors prescribed by q . The extracted parameters are then elementary terms in the remaining source labels. This proves both assertions. $\square$

Proposition 2.5. *Polynomial objects and finite-pattern maps form a small category $\mathcal{E}(\mathbf{P})$.*

Proof. Identities use the undeveloped pattern partition. Given $f : A \rightarrow B$ and $g : B \rightarrow C$, apply Lemma 2.4 to the source cylinders in a description of g . On each resulting source branch of f , the parameters needed by g are elementary terms in the source labels. Substituting them into the output of g gives another liberal pattern: successive successors combine, and

every seed argument remains an X -sorted label. Thus composition is closed. Associativity and the unit laws hold because morphisms are functions.

The objects form a set. Finite descriptions use a set-sized signature and finitely many sorted labels, which may be chosen from fixed countable lists for each sort. They therefore form a set as well, so the category is small. $\square$

2.3 Finite limits

Products will be represented by products of polynomial objects. Equalizers require solving equations between the constructor terms on a source branch. The following solved-form statement also supplies the factor maps needed for their universal property.

Lemma 2.6 (Sorted unification). *Let V be a finite set of sorted variables, and consider a finite system of well-sorted equations between constructor terms in V . Either the system has no solutions or there are a subset $W \subseteq V$ and a tuple of constructor terms $\theta = (\theta_v)_{v \in V}$ in W such that $\theta_w = w$ for $w \in W$ and evaluation gives a bijection*

$$\theta : \prod_{w \in W} S_w \xrightarrow{\sim} \left\{ \text{solutions in } \prod_{v \in V} S_v \right\}.$$

Variables absent from the equations may be retained in W . As a map into $\prod_{v \in V} S_v$, the tuple θ is finite-pattern. The global coordinate projection

$$\pi_W : \prod_{v \in V} S_v \longrightarrow \prod_{w \in W} S_w$$

is finite-pattern and satisfies $\pi_W \theta = \text{id}$ and $\theta \pi_W = \text{id}$ on the solution set.

Proof. Use the usual sorted syntactic unification rules (compare [1, Section 2.2]). Delete an equation $t = t$. For two constructor terms, different outer constructors give failure; equal outer constructors can be cancelled, leaving equations between their corresponding arguments. For an equation $v = t$, with v a variable, eliminate v by substitution if it does not occur in t ; if it occurs properly, declare failure. An equation with a variable on the right is first oriented in this form.

These rules terminate. Keep a set of the original variables not yet eliminated, including those absent from the current equations. Variable elimination decreases the size of this set. Every other nonfailure operation decreases the total number of term symbols in the equations. The lexicographically ordered pair of these nonnegative integers therefore decreases at each operation, with orientation treated as part of the operation to which it applies.

Each rule preserves the stated solution problem. Constructor cancellation and constructor failure are justified by injectivity and disjointness in (2.1). Eliminating $v = t$ identifies solutions with those of the substituted system: the removed coordinate is recovered by evaluating t . The occurs check is sound because a proper occurrence of a variable in a term of its own sort can only have the form $v = s_j^k(v)$ with $k > 0$. This contradicts the height function. No X_i -sorted term has a proper subterm, and no seed accepts a U -sorted argument.

If failure does not occur, record and compose the eliminated substitutions. At termination they express every original variable as a term in the remaining variables W , and leave each $w \in W$ unchanged. The remaining equation list is empty, so any assignment to W extends uniquely to a solution. All sorts are nonempty, hence this case indeed has solutions. The resulting terms have precisely the elementary forms of Definition 2.1. This proves that θ and π_W are finite-pattern and gives their asserted identities. $\square$

Proposition 2.7 (Finite limits). *The category $\mathcal{E}(\mathbf{P})$ has finite limits.*

Proof. The polynomial 1 is terminal. Products are represented by multiplication of polynomials. Their projections are finite-pattern, and the pairing of two finite-pattern maps is finite-pattern after taking a common refinement of their source partitions.

For $f, g : A \rightarrow B$, choose a common source pattern partition (p_r) on which both maps have fixed liberal outputs. Branches whose outputs have different summand tags contribute no solutions. On every other branch, equate the corresponding output terms and apply [Lemma 2.6](#) with $V = L(p_r)$, retaining all source labels. Discard any inconsistent branch. For each remaining branch write W_r and θ_r for its solved form, and let

$$E = \coprod_r \prod_{w \in W_r} S_w.$$

The maps $\iota_{p_r} \theta_r$ define a finite-pattern map $e : E \rightarrow A$. It is injective: each solved form is injective and the source cylinders are disjoint. Its image is exactly the set where $f = g$.

Suppose $h : C \rightarrow A$ satisfies $fh = gh$. By [Lemma 2.4](#), refine its source partition so that each branch maps into a fixed cylinder of p_r and the extracted tuple $\iota_{p_r}^{-1}h$ is elementary in that branch's labels. It takes values in the solution set of the corresponding equations. The required map to the r -th summand of E is therefore

$$\pi_{W_r} \iota_{p_r}^{-1} h.$$

This is a finite-pattern formula on the branch, and $\theta_r \pi_{W_r} = \text{id}$ on solutions shows that its composite with e is h . The formulas define a finite-pattern map $C \rightarrow E$. Its uniqueness follows from injectivity of e . Thus e is an equalizer. Finite products and equalizers give all finite limits. $\square$

Proposition 2.8 (Extensivity). *Finite coproducts in $\mathcal{E}(\mathbf{P})$ are disjoint and stable under pullback. In particular, $\mathcal{E}(\mathbf{P})$ is small and left extensive.*

Proof. Addition of polynomials represents coproducts: the tagged injections are finite-pattern, and finite-pattern maps out of the summands combine by joining their source partitions. The polynomial 0 is initial. Different coproduct tags are disjoint, so distinct injections have pullback 0; each injection is a monomorphism.

Let $f : C \rightarrow A \amalg B$. In a finite-pattern description of f , every source branch lands in a fixed target summand and hence in either A or B . Let C_A be the coproduct of the parameter objects of branches of the first kind, and define C_B similarly. Pattern evaluation and extraction give an isomorphism

$$C_A \amalg C_B \xrightarrow{\sim} C.$$

There are induced finite-pattern maps $C_A \rightarrow A$ and $C_B \rightarrow B$. To check that C_A is the pullback of $A \rightarrow A \amalg B$, take $h : D \rightarrow C$ and $k : D \rightarrow A$ with fh equal to the tagged inclusion of k . Refine h by the preimages of the source cylinders of f . Only branches defining C_A can occur. Their parameter-extraction formulas yield a finite-pattern map $D \rightarrow C_A$ with the required composites, unique because $C_A \rightarrow C$ is injective. The same argument applies to C_B . The empty coproduct is also stable: a map to the empty underlying set has empty source. These decompositions prove pullback stability of finite coproducts. $\square$

3 Isomorphisms and the Burnside semiring

Define the commutative semiring

$$S(\mathbf{P}) = \mathbb{N}[x_i, u_j \mid i \in I, j \in J] / (u_j = u_j + P_j(\mathbf{x}) : j \in J), \quad (3.1)$$

where the quotient is taken by the semiring congruence generated by the displayed relations. For a polynomial object A , write $[A]$ for its class in $S(\mathbf{P})$.

Theorem 3.1 (Object isomorphisms). *For polynomial objects A and B , the following are equivalent:*

- (i) $A \cong B$ in $\mathcal{E}(\mathbf{P})$;
- (ii) $[A] = [B]$ in $S(\mathbf{P})$.

Thus the semiring of isomorphism classes of objects of $\mathcal{E}(\mathbf{P})$, with addition induced by coproduct and multiplication by product, is canonically isomorphic to $S(\mathbf{P})$.

The proof has two inputs. First, the sum of the monomials in the free labels of a pattern partition depends only on its polynomial object (Lemma 3.4). Second, a finite-pattern inverse forces every source label to occur exactly once in each output pattern (Lemma 3.6). The resulting source and target partitions have the same monomials, proving (i) $\Rightarrow$ (ii). The converse follows from the constructor isomorphisms $U_j \cong U_j \amalg D_j$.

3.1 Weights of pattern partitions

Definition 3.2 (Weight). For a linear pattern p , define its weight by

$$\text{wt}(p) = \prod_{\lambda \in L(p)} v_{S_\lambda} \in S(\mathbf{P}), \quad v_{X_i} = x_i, \quad v_{U_j} = u_j.$$

The weight of a finite family of linear patterns is the sum of their weights.

Lemma 3.3 (Development preserves weight). *Developing a frontier U_j -label one step preserves total weight in $S(\mathbf{P})$. Hence every finite development of a linear pattern preserves its weight.*

Proof. Let m be the product of the variables associated to the other labels. The original pattern has weight mu_j . Its successor branch has weight mu_j , while its seed branches have total weight $mP_j(\mathbf{x})$. Their sum is mu_j by (3.1). The assertion for a finite development follows by iteration. $\square$

Lemma 3.4 (Weight of a pattern partition). *If $(p_r)_{r \in F}$ is a finite pattern partition of a polynomial object A , then*

$$\sum_{r \in F} \text{wt}(p_r) = [A] \quad \text{in } S(\mathbf{P}).$$

Proof. Choose N strictly larger than every successor depth appearing in the patterns p_r . In a coordinate of the form $s_j^r(\lambda)$, develop the frontier label λ to depth $N - r$. This replaces that coordinate by the tail $s_j^N(\mu)$ and the seed patterns at depths $r, r + 1, \dots, N - 1$. A coordinate already of the form $s_j^r c_{j,a}(\lambda_{i,k})$, with $r < N$, needs no further development. Coordinates of sort X_i are unchanged.

Taking these developments in every coordinate refines each p_r into members of the standard depth- N partition of A , up to renaming labels. Every such standard cylinder

is nonempty. Since the original cylinders are disjoint and cover $|A|$, the refined families together contain each standard cylinder exactly once. By Lemma 3.3, their total weight equals $\sum_r \text{wt}(p_r)$. The same standard partition is a development of the undeveloped monomial summands of A , whose total weight is $|A|$. Another application of Lemma 3.3 proves the identity. For $A = 0$, both families are empty and both weights are zero. $\square$

3.2 Linear output patterns for isomorphisms

An injective finite-pattern map cannot omit a source label. To exclude repetition, we use the inverse map: it expresses the image of each source cylinder as a finite union of linear cylinders. The next lemma shows why repetition is incompatible with such an image.

Lemma 3.5 (Repeated labels). *Let L be a finite set of sorted labels, and let q be a liberal pattern over L in which a label λ occurs at least twice. Fix values of all labels in $L \setminus \{\lambda\}$, and let Γ be the subset of the target obtained by varying λ through its sort. Then Γ is infinite. If a linear cylinder D is contained in the image of q , there is a finite development of D such that each refined cylinder meets Γ in at most one point. Consequently, the image of q is not a finite union of linear cylinders.*

Proof. Every sort is infinite: this is part of the choice of X_i , and $U_j = \mathbb{N} \times D_j$ is infinite because $P_j \neq 0$ implies $D_j \neq \emptyset$. Evaluation of a constructor term is injective in each label that occurs in it, with the other labels held fixed. Thus varying λ gives infinitely many distinct points of Γ .

Choose two occurrences of λ in q . Each determines an extraction from the target: select its product coordinate, cancel the prescribed successors, and, if the occurrence is a seed argument, select that argument. Both extractions are defined on the image of q , and both return the value assigned to λ .

Let D be a linear cylinder contained in this image. Develop its frontier U -labels far enough to decide all constructor tests along the two extraction paths. This requires only finitely many successor cancellations and seed tests, so it gives a finite development. On every refined cylinder D' , the extracted values are constructor terms r_1, r_2 in its frontier labels.

These two terms use disjoint sets of frontier labels. Indeed, if λ has sort U_j , its two occurrences lie in distinct product coordinates: a seed constructor has no U -sorted arguments. If λ has sort X_i , the occurrences either lie in distinct product coordinates or occupy different argument slots of the same seed constructor. Distinct coordinates of a linear pattern use disjoint labels. After a seed is exposed, its distinct argument slots also use distinct labels. Cancelling successors and selecting arguments therefore leaves disjoint sets of labels in both cases.

Since $D' \subseteq \text{im}(q)$, we have

$$r_1(\rho) = r_2(\rho)$$

for every assignment ρ to the frontier labels of D' . If r_1 contained a label, we could fix all other labels and vary that one. Injectivity of the constructors would change r_1 , whereas the disjointness just proved would keep r_2 fixed. This is impossible. The same argument applies to r_2 , so both terms are ground, meaning that they contain no labels.

There are no ground terms of sort X_i . Thus, in that case, no nonempty D' as above can occur. A ground term of sort U_j can occur, but must have the form $s_j^n c_{j,a}()$ for a nullary seed constructor. In this case the common extracted value is fixed on D' . At a point of Γ this value is exactly the value of λ ; hence at most one value of λ can occur in $D' \cap \Gamma$. The other source labels were fixed when Γ was defined, so this intersection has at most one point.

It follows that $D \cap \Gamma$ is finite. A finite union of linear cylinders contained in $\text{im}(q)$ therefore meets Γ in a finite set, and cannot contain the infinite subset $\Gamma \subseteq \text{im}(q)$. $\square$

Lemma 3.6 (Linear patterns for an isomorphism). *Let $f : A \rightarrow B$ be an isomorphism in $\mathcal{E}(\mathbf{P})$. Then f has a finite-pattern description $(p_r, q_r)_{r \in F}$ such that:*

- (i) $(p_r)_{r \in F}$ is a pattern partition of A ;
- (ii) $(q_r)_{r \in F}$ is a pattern partition of B ;
- (iii) for each r , the patterns p_r and q_r have exactly the same sorted labels, each occurring once in each pattern;
- (iv) on each branch, f applies the same substitution to p_r and q_r .

Proof. Start with a finite-pattern description (p_r, q_r) of f . No label of p_r can be omitted from q_r : changing only an omitted label would give distinct points of the source cylinder with the same image, contrary to injectivity of f .

Let $g : B \rightarrow A$ be the inverse morphism. For each source cylinder $C = \text{im}(\iota_{p_r})$, we have

$$f(C) = g^{-1}(C). \quad (3.2)$$

By Lemma 2.4, the right-hand side is a finite union of linear cylinders in B . The left-hand side is the image of q_r . If a label occurred more than once in q_r , this would contradict Lemma 3.5. Hence each source label occurs exactly once in q_r , and q_r is linear.

The cylinders of the q_r are pairwise disjoint by injectivity of f , and they cover $|B|$ by surjectivity. Thus they form the required pattern partition; the branchwise substitution rule is part of the original description of f . $\square$

Proof of Theorem 3.1. Suppose that $A \cong B$. By Lemma 3.6, choose pattern partitions (p_r) of A and (q_r) of B with the same sorted labels on corresponding branches. Then $\text{wt}(p_r) = \text{wt}(q_r)$ for every r . Applying Lemma 3.4 to the two partitions gives

$$[A] = \sum_r \text{wt}(p_r) = \sum_r \text{wt}(q_r) = [B] \quad \text{in } S(\mathbf{P}).$$

Conversely, the relation of being isomorphic in $\mathcal{E}(\mathbf{P})$ is a semiring congruence on polynomial objects. Indeed, coproducts and products preserve isomorphisms, and the canonical associativity, symmetry, unit, and distributivity maps are finite-pattern maps with finite-pattern inverses. For each j , the constructor decomposition (2.1) gives an isomorphism

$$U_j \xrightarrow{\sim} U_j \amalg D_j.$$

It sends $s_j(u)$ to u in the U_j -summand, and sends $c_{j,a}(d)$ to d in the corresponding summand of D_j ; the inverse uses s_j and the $c_{j,a}$. Both maps are finite-pattern. Thus the isomorphism congruence contains every relation $u_j = u_j + P_j(\mathbf{x})$ and hence the congruence they generate. Equality in $S(\mathbf{P})$ implies $A \cong B$, proving the converse and the asserted identification of semirings. $\square$

3.3 Group completion

Corollary 3.7. *The Grothendieck ring with respect to coproducts and products is*

$$K_0^\amalg(\mathcal{E}(\mathbf{P})) \cong \mathbb{Z}[x_i, u_j \mid i \in I, j \in J] / (P_j(\mathbf{x}) \mid j \in J).$$

Proof. By Theorem 3.1, the semiring of object classes is $S(\mathbf{P})$. The additive group completion of a commutative semiring is the universal ring receiving a semiring homomorphism from it. For any commutative ring T , a semiring homomorphism $S(\mathbf{P}) \rightarrow T$ is exactly a choice of images for the x_i and u_j satisfying

$$u_j = u_j + P_j(\mathbf{x}), \quad \text{equivalently} \quad P_j(\mathbf{x}) = 0.$$

The polynomial-ring quotient in the statement represents precisely these choices, so it has the required universal property. $\square$

4 Additive spans and categorical dimension

Let $\mathcal{E}$ be a small lextensive category. A span from X to Y is a diagram

$$X \xleftarrow{p} M \xrightarrow{q} Y,$$

or equivalently an object of the slice $\mathcal{E}/(X \times Y)$. Isomorphisms of spans are isomorphisms of their middle objects commuting with both legs. Their isomorphism classes form a commutative monoid under coproduct of middle objects.

Definition 4.1. The *additive span category* $\mathcal{A}(\mathcal{E})$ has the same objects as $\mathcal{E}$ and morphism groups

$$\mathrm{Hom}_{\mathcal{A}(\mathcal{E})}(X, Y) = K_0^{\mathrm{II}}(\mathcal{E}/(X \times Y)).$$

On spans, composition is defined by pullback:

$$(X \leftarrow M \rightarrow Y), \quad (Y \leftarrow N \rightarrow Z) \quad \longmapsto \quad (X \leftarrow M \times_Y N \rightarrow Z),$$

and is extended bilinearly to the group completions.

Pullback preserves finite coproducts in $\mathcal{E}$, so this composition is additive in each variable before group completion and the bilinear extension is well-defined. Associativity follows from the canonical isomorphism between the two iterated pullbacks. The identity of X is the span $X \leftarrow X \rightarrow X$ with both legs equal to the identity. The slice categories are small, hence so is $\mathcal{A}(\mathcal{E})$.

Proposition 4.2. *The category $\mathcal{A}(\mathcal{E})$ is additive and rigid symmetric monoidal. Its tensor product is induced by Cartesian product in $\mathcal{E}$, its tensor unit is 1, and every object is self-dual.*

Proof. The morphism groups and bilinear composition make the category preadditive. The initial object 0 of $\mathcal{E}$ is strict, so every span with source or target 0 has initial middle object; thus 0 is a zero object of $\mathcal{A}(\mathcal{E})$. For $X \amalg Y$, define the injection and projection associated to X by

$$i_X = [X \xleftarrow{\mathrm{id}} X \xrightarrow{\iota_X} X \amalg Y], \quad p_X = [X \amalg Y \xleftarrow{\iota_X} X \xrightarrow{\mathrm{id}} X],$$

and use the analogous spans for Y . Disjointness of coproducts gives

$$p_X i_X = \mathrm{id}_X, \quad p_Y i_Y = \mathrm{id}_Y, \quad p_X i_Y = 0, \quad p_Y i_X = 0.$$

The coproduct of the spans representing $i_X p_X$ and $i_Y p_Y$ is the identity span on $X \amalg Y$. Hence $i_X p_X + i_Y p_Y = \mathrm{id}_{X \amalg Y}$, which proves that $X \amalg Y$ is a biproduct.

Cartesian product of spans respects composition: the product of two pullback diagrams is a pullback diagram. The associativity, unit, and symmetry isomorphisms come from those in $\mathcal{E}$. For every Z , pullback stability of coproducts gives

$$Z \times (X \amalg Y) \cong (Z \times X) \amalg (Z \times Y).$$

Consequently the product of spans is additive in both variables and extends to a biadditive tensor product on the group completions.

Take $X^\vee = X$, with coevaluation and evaluation

$$\text{coev}_X = [1 \xleftarrow{!} X \xrightarrow{\Delta} X \times X], \quad \text{ev}_X = [X \times X \xleftarrow{\Delta} X \xrightarrow{!} 1].$$

In the composite

$$X \xrightarrow{\text{coev}_X \otimes \text{id}_X} X \times X \times X \xrightarrow{\text{id}_X \otimes \text{ev}_X} X,$$

the pullback identifies triples of the forms (a, a, b) and (c, d, d) . Its four coordinates are therefore all equal, so the middle object is canonically X and both outer legs are the identity. The other zig-zag identity follows from the same pullback with the factors interchanged. Thus X is its own dual. $\square$

The functor from $\mathcal{E}$ to $\mathcal{A}(\mathcal{E})$ sending $f : X \rightarrow Y$ to $[X \xleftarrow{\text{id}} X \xrightarrow{f} Y]$ sends finite coproducts to biproducts and Cartesian products to tensor products. It induces a surjective ring map

$$\eta : K_0^\Pi(\mathcal{E}) \longrightarrow K_0^\oplus(\mathcal{A}(\mathcal{E})), \quad [X] \longmapsto [X], \quad (4.1)$$

since the two categories have the same objects. To prove injectivity, we use the dimension of an object to recover its original Burnside class.

Proposition 4.3. *The map (4.1) is an isomorphism. Under the identification*

$$\text{End}_{\mathcal{A}(\mathcal{E})}(1) \cong K_0^\Pi(\mathcal{E}),$$

its inverse is induced by categorical dimension.

Proof. A span from 1 to itself is specified by its middle object. Coproduct of middle objects is addition, and composition is their Cartesian product. Thus

$$\text{End}_{\mathcal{A}(\mathcal{E})}(1) = K_0^\Pi(\mathcal{E}/(1 \times 1)) \cong K_0^\Pi(\mathcal{E})$$

as rings. Using the duality in [Proposition 4.2](#), the trace of id_X is the span obtained by pulling the diagonal back along itself. Since the diagonal is a monomorphism, its middle object is X , and

$$\dim(X) = [1 \xleftarrow{!} X \xrightarrow{!} 1] = [X].$$

Categorical trace is invariant under conjugation by an isomorphism. It follows that this class depends only on the isomorphism class of X in $\mathcal{A}(\mathcal{E})$. Moreover the displayed formula gives

$$\dim(X \oplus Y) = [X] + [Y], \quad \dim(X \otimes Y) = [X][Y], \quad \dim(1) = 1.$$

It therefore induces a ring map

$$\delta : K_0^\oplus(\mathcal{A}(\mathcal{E})) \longrightarrow K_0^\Pi(\mathcal{E})$$

with $\delta\eta = \text{id}$. Since η is surjective, it is an isomorphism with inverse δ . $\square$

Proof of Theorem 1.2. By [Propositions 2.7](#) and [2.8](#), the category $\mathcal{E}(\mathbf{P})$ is small and lexextensive. Set $\mathcal{A}_{\mathbf{P}} = \mathcal{A}(\mathcal{E}(\mathbf{P}))$. It is additive rigid symmetric monoidal, with every object self-dual, by [Proposition 4.2](#). Finally, [Corollary 3.7](#) and [Proposition 4.3](#) give

$$K_0^\oplus(\mathcal{A}_{\mathbf{P}}) \cong K_0^\Pi(\mathcal{E}(\mathbf{P})) \cong \mathbb{Z}[x_i, u_j \mid i \in I, j \in J]/(P_j \mid j \in J).$$

$\square$

5 Realizing an arbitrary ring

5.1 Nonnegative relations and an object of class -1

Let $R \neq 0$ be a commutative unital ring. Choose a presentation

$$R \cong \mathbb{Z}[x_i \mid i \in I] / (f_j \mid j \in J) \quad (5.1)$$

with every $f_j \neq 0$. Such a presentation is obtained by taking one generator for each element of R and all nonzero elements of the kernel as relations. Write

$$f_j = f_j^+ - f_j^-, \quad f_j^+, f_j^- \in \mathbb{N}[x_i \mid i \in I],$$

where the two polynomials have disjoint supports. Adjoin an indeterminate z and a new relation index $0 \notin J$, and put

$$P_0 = 1 + z, \quad P_j = f_j^+ + z f_j^- \quad (j \in J).$$

These are nonzero polynomials with nonnegative coefficients. Applying [Theorem 1.2](#) to the variables x_i, z and relations indexed by $\{0\} \amalg J$ gives a small additive rigid symmetric monoidal category $\mathcal{A}$, with self-dual objects, and

$$\begin{aligned} K_0^\oplus(\mathcal{A}) &\cong \mathbb{Z}[x_i, z, u_k \mid i \in I, k \in \{0\} \amalg J] / (1 + z, f_j^+ + z f_j^- \mid j \in J) \\ &\cong R[u_k \mid k \in \{0\} \amalg J]. \end{aligned} \quad (5.2)$$

The second isomorphism substitutes $z = -1$.

Let Z be the object corresponding to z . Under (5.2),

$$[Z] = -[1]. \quad (5.3)$$

Thus every object W has an actual object representing the negative of its class:

$$[Z \otimes W] = -[W]. \quad (5.4)$$

5.2 The full subcategory of constant classes

Use (5.2) to view R as the constant subring of $K_0^\oplus(\mathcal{A})$. Let $\mathcal{C}_R$ be the full subcategory on the objects X satisfying

$$[X] \in R \subseteq R[u_k \mid k \in \{0\} \amalg J].$$

It contains $0, 1$ and is closed under finite direct sums and tensor products. Since the chosen dual of each object is that object itself, the evaluation and coevaluation maps restrict to $\mathcal{C}_R$. Consequently $\mathcal{C}_R$ is a small additive rigid symmetric monoidal category.

To identify its Grothendieck ring, we must both represent the elements of R and realize within $\mathcal{C}_R$ the stabilizations that identify their representatives in $\mathcal{A}$.

Lemma 5.1. *Every element of R is represented by an object of $\mathcal{C}_R$.*

Proof. Write an element $r \in R$ as $[A] - [B]$ in $K_0^\oplus(\mathcal{A})$. Equation (5.4) gives

$$r = [A \oplus (Z \otimes B)].$$

By definition, the object on the right belongs to $\mathcal{C}_R$. □

Lemma 5.2. *The inclusion induces an injective ring map*

$$K_0^\oplus(\mathcal{C}_R) \longrightarrow K_0^\oplus(\mathcal{A})$$

whose image is R .

Proof. The image is contained in R by definition and contains R by Lemma 5.1. For injectivity, every element of $K_0^\oplus(\mathcal{C}_R)$ is a difference $[A] - [B]$, so suppose that $A, B \in \mathcal{C}_R$ have equal classes in $K_0^\oplus(\mathcal{A})$. Equality in the group completion of a commutative monoid means equality after adding the same monoid element. Hence there exists $W \in \mathcal{A}$ with

$$A \oplus W \cong B \oplus W.$$

The object W need not belong to $\mathcal{C}_R$. Instead, take

$$T = W \oplus (Z \otimes W).$$

Its class is zero by (5.4), so $T \in \mathcal{C}_R$. Adding $Z \otimes W$ to the preceding isomorphism yields

$$A \oplus T \cong B \oplus T.$$

Both sides and T lie in $\mathcal{C}_R$, and fullness makes this an isomorphism in $\mathcal{C}_R$. Thus $[A] = [B]$ in $K_0^\oplus(\mathcal{C}_R)$, proving injectivity. $\square$

Proof of Theorem 1.1. If $R \neq 0$, the category $\mathcal{C}_R$ just constructed is small, additive, rigid, and symmetric monoidal, with self-dual objects. By Lemma 5.2, its split Grothendieck ring is R . The zero ring is realized by the zero category, with its unique symmetric monoidal structure. $\square$

6 The field $\mathbb{F}_4$ and a nonzero nilpotent trace

The smallest nonprime finite field gives an explicit instance of the construction. Take

$$P_1 = 2, \quad P_2 = 1 + x + x^2.$$

The category $\mathcal{A}_{\mathbf{P}}$ of Theorem 1.2 satisfies

$$K_0^\oplus(\mathcal{A}_{\mathbf{P}}) \cong \mathbb{Z}[x, u_1, u_2]/(2, 1 + x + x^2) \cong \mathbb{F}_4[u_1, u_2].$$

Every class has an actual object representative, since $[A] - [B] = [A \oplus B]$ in characteristic two. The full subcategory of objects whose classes are constant has split Grothendieck ring $\mathbb{F}_4$: in the proof of Lemma 5.2, one may take $T = W \oplus W$, whose class is zero.

By Proposition 4.3, the scalar ring in the ambient category is

$$\text{End}_{\mathcal{A}_{\mathbf{P}}}(1) \cong \mathbb{F}_4[u_1, u_2].$$

The same scalar ring is retained in the full subcategory. Its characteristic is two, so scalar action gives $2\text{id}_Y = 0$ for every object Y . Let X be the object corresponding to x , and write $\alpha = [X] \in \mathbb{F}_4$. For the symmetry $\tau : X \otimes X \rightarrow X \otimes X$, set

$$n = \text{id}_{X \otimes X} - \tau.$$

Since $\tau^2 = \text{id}$,

$$n^2 = \text{id} - 2\tau + \tau^2 = 0.$$

The trace of the symmetry is $\dim(X) = \alpha$. In spans this can be seen directly: closing the symmetry identifies its two coordinates, leaving the middle object X . Therefore

$$\text{tr}(n) = \dim(X)^2 - \dim(X) = \alpha^2 - \alpha = 1.$$

Thus this category is not trace-zero in the sense of [3, Definition 3]. The positive-characteristic restriction in [3, Theorem 4 and the paragraph following it] assumes the trace-zero condition, and hence does not apply to this example.

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